Learn-Design-Implement (LDI) Project Structure and Suggestions

Contents

Learn-Design-Implement (LDI) Project Structure and Suggestions 1
Project Structure and Requirements 1
Project Options 2
Self-proposed Projects 3
PhD Students 3

ELEN90099 LDI project aims to achieve the objectives and ILOs listed in the manual. We need to balance these against your "individualised learning." Therefore, we impose a specific structure to ELEN90099 LDI projects while giving you as much freedom as possible. We will provide you an "example" project along with many resources and options so that you make most of ELEN90099 LDI project in your own way.

Project Structure and Requirements

Each project should have the following structure and satisfy the requirements listed below.

1. Choose an engineering domain: e.g. communications, power systems, robotics, etc.
2. Choose a specific engineering system in that domain, e.g. wireless comms, solar PV, robotic arm, ...
3. Define an engineering problem, e.g. traffic analysis, solar output forecasting, robotic arm predictive maintenance
4. Find (or generate) datasets related to that system and problem
5. **Analyse and solve the problem using deep learning methods**.
6. [*optionally*] Use foundation models (LLMs) in conjunction with deep learning.
7. Store your results in a database, preferably use vector database options
8. (If time permits compare to baseline methods from foundation models or classical machine learning.)

Project Options

There is a rich variety of choices based on your background, course, and interests. Here are a few options from different engineering domains.

- You don't have to use one of the suggested datasets; you can use different ones! In many cases, there are multiple alternative suggestions, so choose the one you like.
- *You can and should* further articulate and scope the suggested problems.
- Consider alternative methods from foundation models or classical machine learning to compare against your deep learning solution.

1. Mobile Phone Activity in a City (Communications)

- Analyse and forecast call, SMS, and internet activity in a city area using deep learning.
- The dataset divides the city into a rectangular grid. A subset of the grid could be selected to make the problem size more manageable. See
  https://www.kaggle.com/datasets/marcodena/mobile-phone-activity/data
  https://github.com/arunasubbiah/milan-telecom-data-modeling

2. Solar PV forecasting and battery optimisation (Power Systems)

- Analyse and forecast Solar PV generation data such as this one using various deep learning methods
  https://www.kaggle.com/datasets/pythonafroz/daily-power-production-data-of-solar-power-plant
- Analyse and forecast electricity prices using deep learning time-series forecasting
  https://www.kaggle.com/datasets/aramacus/electricity-demand-in-victoria-australia
  https://www.aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/aggregated-data

3. Fault detection and predictive maintenance (Manufacturing)

- Use deep learning methods to do predictive maintenance and predict machine failures
  https://www.kaggle.com/datasets/stephanmatzka/predictive-maintenance-dataset-ai4i-2020
  https://archive.ics.uci.edu/dataset/963/ur3+cobotops
  https://github.com/boschresearch/CNC_Machining
- Predict remaining useful life (RUL)
  https://www.kaggle.com/datasets/prognosticshse/preventive-to-predicitve-maintenance

4. Cybersecurity and Network Security (Security, Communications)

- Use deep learning methods to detect anomalies and/or specific attacks
  https://archive.ics.uci.edu/dataset/942/rt-iot2022
  https://www.kaggle.com/datasets/emilynack/aci-iot-network-traffic-dataset-2023/data
- Investigate adversarial machine learning with attacks to ML models and adversarial training as a counter measure
  https://github.com/Trusted-AI/adversarial-robustness-toolbox
  https://github.com/andreaventuri01/DReLAB_tutorial

5. Time Series forecasting – State of the Art Methods (Engineering)

- Study and implement state of the art time series forecast models such as transformers and modern RNN variants
  https://github.com/thuml/Time-Series-Library
- Apply these methods to engineering datasets such as web traffic, electricity demand, smart meter data
  https://www.kaggle.com/datasets/konradb/monash-time-series-forecasting-repository
- Compare your results to LSTM, GRU, ARMAX baselines and provide a detailed analysis
- If time permits, try major available foundation models as another set of comparison

Self-proposed Projects

Self-proposed projects are allowed only if they are explicitly approved by the lecturer (not the demonstrator). As you can expect, the lecturer will try to make sure that your creative project somehow incorporates the aspects listed in the requirements because the idea is for you to get hands-on experience with those. Please keep this in mind when proposing projects.

PhD Students

Formally enrolled PhD students can (maybe should) align the LDI project with their research topics as much as possible. All PhD students must talk to the lecturer about the LDI project before starting.